MECH 325 Machine Elements

Gear Trains

Tutorial 4 – Helical and Bevel Gear Stress Analysis

1. Adapted from **Shigley, Problem 14-22**. (10th ed 14-19)

   A commercial enclosed gear drive is to be designed. The drive is to consist of two helical gears with a 20° pressure angle and a helix angle of 25°. The pinion is to have 16 teeth driving a 48- tooth gear. The pinion speed is 300 rev/min, the face width is 2 in, and a normal diametrical pitch of 6 teeth/in. The gears are Grade 1 steel made to No. 6 quality standards. The gears will be uncrowned, centered on their shafts between bearings, and are to be accurately and rigidly mounted. Assume a pinion life of 10^8 cycles and a reliability of 0.90. The power to be transmitted is 5 hp.

   Part 1: To be demonstrated by the TA.

   a. Determine the AGMA bending stress in the pinion
   b. Determine the minimum AGMA bending strength number for the through-hardened pinion
   c. If a Grade 1 through-hardened at 200 Brinell gear is to be used what will be the corresponding safety factor for the pinion bending stress
   d. Determine the AGMA contact stress in the pinion

   Part 2: To be done by the student

   e. Determine the minimum AGMA contact strength number for the through-hardened pinion
   f. If a Grade 1 through-hardened at 200 Brinell gear is to be used what will be the corresponding safety factor for the pinion contact stress

   Part 3: To be demonstrated by the TA

   g. Select a suitable steel alloy and heat treatment to meet this requirement. What is the final safety factor

   Part 4: Extra practice, to be done by students at home.

2. Adapted from **Shigley, Problem 15-1**.

An uncrowned straight-bevel pinion has 20 teeth, a diametrical pitch of 6 teeth per inch and a transmission accuracy of 6. Both the pinion and gear are made of through-hardened steel with a Brinell hardness of 300. The driven gear has 60 teeth. The gearset has a life goal of 10^9 revolutions of the pinion with a reliability of 0.999. The shaft angle is 90° and the pinion speed is 900rpm. The face width is 1.25 inches, and the normal pressure angle is 20°. The pinion is mounted outboard of the bearings (cantilevered) while the gear is straddle mounted. Based on the AGMA bending strength, what is the power rating of the gearset. Assume overload factor is 1.0 and S = S = 1.0